

Generation Interconnection Request System Impact Study Report

For Queue Project Q11-0058

North Tifton 100 kV
October 01, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0058. The Interconnection Customer proposes a 373.3 MW natural gas combined cycle facility interconnecting at the North Tifton 100 kV substation in the Indiana area. The request was received on July 17, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 43 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$47,062,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: WEST ASHEVILLE / N TIFTON 100kV Line
Contingency: N TIFTON / WARTBURG 100kV L
Loading: 137.4% (Rating: 115 MVA)
DFAX: 32.0%
Pre-project loading: 33.3%
2. Facility: OAK PLAINS / N TIFTON 100kV
Contingency: WEST FORD FLAT / NORTH TIFTON 100kV line
Loading: 128.5% (Rating: 115 MVA)
DFAX: 51.3%
Pre-project loading: 37.6%
3. Facility: WARTBURG / W ASHEVILLE 100kV
Contingency: WEST ASHEVILLE / N TIFTON 100kV
Loading: 125.0% (Rating: 100 MVA)
DFAX: 27.7%
Pre-project loading: 21.6%
4. Facility: N TIFTON SUBSTATION / WARTBURG 100kV
Contingency: N TIFTON / WEST ASHEVILLE 100kV Line
Loading: 119.1% (Rating: 117 MVA)
DFAX: 27.7%
Pre-project loading: 30.9%
5. Facility: WAYNEDALE / MONFORT 230kV
Contingency: MONTPELIER ELECTRIC GENERATING STATION / WABASH SUBSTATION 500kV
Loading: 111.2% (Rating: 700 MVA)
DFAX: 6.8%
Pre-project loading: 107.6%

6. Facility: STATE ST / WAYNE DALE 230kV
 Contingency: MUSKOGEE / MONTPELIER ELECTRIC GENERATING STATION 500kV
 Loading: 107.0% (Rating: 572 MVA)
 DFAX: 10.5%
 Pre-project loading: 100.1%
7. Facility: N TIFTON / WEST FORD FLAT 100kV
 Contingency: OAK PLAINS / NORTH TIFTON 100kV
 Loading: 100.3% (Rating: 127 MVA)
 DFAX: 47.8%
 Pre-project loading: 40.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WEST ASHEVILLE / N TIFTON 100kV Line	Rebuild line with higher-capacity conductor	\$9,263,000	100% (\$9,263,000)
OAK PLAINS SUBSTATION / N TIFTON 100kV	Rebuild line with higher-capacity conductor	\$8,875,000	100% (\$8,875,000)
WARTBURG / W ASHEVILLE 100kV	Rebuild line with higher-capacity conductor	\$8,684,000	100% (\$8,684,000)
N TIFTON SUBSTATION / WARTBURG 100kV	Rebuild line with higher-capacity conductor	\$6,305,000	100% (\$6,305,000)
WAYNE DALE / MONFORT 230kV	Rebuild line with higher-capacity conductor	\$52,169,000	16% (\$8,109,000)
STATE ST / WAYNE DALE 230kV	Reconductor line	\$6,309,000	24% (\$1,530,000)
NORTH TIFTON / WEST FORD FLAT SUBSTATION 100kV	Reconductor line	\$4,296,000	100% (\$4,296,000)
	Total allocated to project		\$47,062,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.